

École Numérique

Gestion Scolaire — Burkina Faso

Technical Documentation

This document describes the architecture, features, and functional modules of the **École Numérique** school management web application — a self-contained, offline-first HTML system designed for primary schools in Burkina Faso.

File	Ecole-Main-System.html
Language	French (Interface) / JavaScript (Logic)
Lines of Code	~8,336
Technology	HTML5 · CSS3 · Vanilla JavaScript · Chart.js
Storage	Browser localStorage (no server required)
Target Users	Admin, Teachers, Parents, Students

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1. System Overview

École Numérique is a **self-contained, single-page web application** for managing a primary school in Burkina Faso. The entire system — UI, logic, and data model — lives in one HTML file. It requires no server, no database, and no internet connection after the initial load of Google Fonts and Chart.js.

All data is persisted in the **browser's localStorage**, making it immediately usable offline on any modern device. The design palette draws on West African earth tones (terracotta, ochre, savanna, earth) for a culturally resonant interface.

Key Characteristics

- **Zero-dependency deployment** — Open the HTML file in any browser; no installation needed.
- **Role-based access** — Four distinct portals: Admin, Teacher, Parent, and Student.
- **French-language UI** — Fully localised to the Burkinabè school system.
- **Burkina academic calendar** — Three trimesters: T1 Oct–Dec, T2 Jan–Mar, T3 Apr–Jun.
- **CEP examination support** — Tracks Certificat d'Études Primaires results for CM2 pupils.
- **Offline-first** — Uses localStorage; no network calls for core functionality.

2. Authentication & Login

The app opens on a login screen with a role selector and username/password fields. Each role unlocks a different navigation menu and feature set:

Role	Default Username	Access Level
Administrateur	admin	Full access to all modules
Enseignant (Teacher)	teacher1	Students, grades, absences, homework, messages
Parent	parent1	Absences, homework, payments, bulletins, messages
Élève (Student)	student1	Own grades, absences, homework, schedule

On successful login the login card hides and the main app container appears. The sidebar is dynamically built from a **navigationConfig** object that maps each role to its permitted sections.

3. Application Layout

After login the interface splits into two areas:

- **Sidebar (280 px)** — Fixed left panel with the school logo, logged-in user badge, role-specific navigation links, and a logout button. The active item is highlighted with a terracotta-to-ochre gradient.
- **Main content area** — Scrollable right panel showing the active section. Only one section is visible at a time; switching is handled by toggling CSS display properties.

4. Functional Modules

4.1 Tableau de Bord (Dashboard)

The dashboard is role-aware and shows different KPI cards depending on who is logged in.

- **Admin:** Four stat cards (total students, teachers, classes, payments collected) plus three Chart.js charts: class averages by subject (bar), CEP results by gender (pie), and a dual-axis line chart of CEP pass rate and school average over 2022–2024.
- **Teacher:** Cards for total students and classes, plus quick links to absences and grades.
- **Parent:** Child's attendance summary and pending payment alerts.
- **Student:** Personal attendance count and subject average summary.

4.2 Gestion des Élèves (Student Registration)

- Search and filter by name, class, or matricule.
- Add / edit / archive students — name, sex, date of birth, guardian info, class assignment.
- Automatic matricule generation.
- Promotion workflow to advance students to the next level at year-end.
- Archived (graduated/transferred) students are moved to a separate archive section.
- Student health records: allergies, blood type, vaccinations, emergency contacts.

4.3 Suivi des Absences (Absence Tracking)

- Absence entry form: student, date, reason (justified / unjustified), half-day or full day.
- Absence history table with delete capability for admins and teachers.
- Parents and students see a read-only view filtered to their child / self.
- Count summaries displayed on the dashboard.

4.4 Notes et Moyennes (Grades & Averages)

Three sub-tabs cover the complete grading workflow:

Tests tab

- Enter individual test scores per student, subject, trimester, and type (Devoir / Composition).
- Scores stored as `testScores[studentId][subjectId][trimester][array of entries]`.
- Filter by class and student; edit or delete existing scores.

Moyennes (Averages) tab

- Select a student and trimester to view per-subject averages computed on the fly.
- Displays individual test scores, average per subject, and an overall general average.
- Colour-coded badges (green ≥ 10 , red < 10) indicate pass / fail.

Résultats CEP tab (Admin only)

- Lists all CM2 students with a toggle to mark each as admitted to the CEP exam.
- Summary bar: total CM2 count, number admitted, and success rate percentage.
- CEP data feeds directly into the dashboard pie chart.

4.5 Bulletin Scolaire (Report Cards)

- School header: logo, name, address, and academic year.
- Student identity block: name, matricule, class, trimester.
- Subject-by-subject table: test scores, computed average, rank mention.
- Overall general average and class rank.
- Headmaster and économe (bursar) signatures uploaded as images in Configuration.
- Print button triggers the browser print dialog for the bulletin panel.

4.6 Emploi du Temps (Timetable)

A weekly grid (Monday–Friday, morning/afternoon slots) editable by the admin. Teachers see only their assigned class schedules. Data is stored per class in `appState.schedule`.

4.7 Gestion des Paiements (Payments)

- Record payments per student: amount, date, trimester, payment method.
- Configure class-level fee plans (monthly, trimester, annual) with due dates.
- Fee exemptions: grant partial or full exemption with justification.
- Payment history table with CSV export.
- Économe signature printed on payment receipts.
- Dashboard total shows cumulative fees collected.

4.8 Devoirs (Homework)

Teachers create homework assignments per class and subject. Parents and students view pending and past assignments. The module tracks:

- Title, description, due date, subject, and class.
- Submission status per student (submitted / not submitted).
- Optional grading and written feedback after submission.

4.9 Communications & Messagerie

- Internal messaging between users, filterable by role.
- Announcements board: admin posts school-wide notices targeting specific roles.
- Inbox and sent folders with unread message counter.
- Discipline records: teachers log behavioural incidents with a severity level.
- Merit points system: award positive recognition points to students.

4.10 Rapports (Reports & Exports)

- Export full student list to CSV.
- Export all grades and averages by trimester to CSV.
- Export payment records to CSV.
- Parents can print their child's bulletin via the browser print dialog.

4.11 Registration Enseignant (Teacher Management)

- Add / edit / deactivate teacher profiles.
- Assign teachers to classes and subjects via the teacherAssignments map.
- Leave requests: submit, approve, or reject leave with substitute assignment.

4.12 Archives

Graduated or transferred students are archived rather than deleted. The archive section lets the admin browse former students, view their records, and restore them if needed.

4.13 Configuration (Admin Settings)

- School information: name, address, phone, email, academic year, headmaster name.
- School logo upload (file or URL) — appears on bulletins and watermarks.
- Headmaster and économe (bursar) signature image uploads.
- Subject management: add, rename, or remove subjects across all classes.
- Class management: add or edit class levels (CP1, CP2, CE1, CE2, CM1, CM2).
- Academic year start date for automatic trimester detection.

5. Data Model (appState)

All application state lives in a single JavaScript object called **appState**, serialised to localStorage under the key "ecoleData" on every mutation. The main collections are shown below.

Key	Type	Description
users	Array	User accounts: role, username, password, linked studentId/classId
classes	Array	School classes: id, name, level (CP1–CM2), capacity
students	Array	Student profiles: name, sex, DOB, guardian, classId, matricule
grades	Object	{studentId: {subjectId: {trimester: score}}}
testScores	Object	{studentId: {subjectId: {trimester: [{name, score, maxScore, date, type}]}}}
absences	Array	Absence records: studentId, date, reason, type (half/full day)
schedule	Object	{classId: {day: {slot: subject}}}
payments	Array	Payment records: studentId, amount, date, trimester, method
classFees	Object	{classId: amount}
paymentPlans	Object	{classId: {type, amount, dueDate}}
feeExemptions	Object	{studentId: {reason, percentage, approvedBy, date}}
cepResults	Object	{studentId: "oui" "non"}
messages	Array	Internal messages: from, to, subject, body, date, read flag
announcements	Array	School-wide notices with targetRole
teacherAssignments	Object	{teacherId: [{classId, subjectId}]}
disciplineRecords	Array	Behavioural incidents with severity level
meritPoints	Object	{studentId: points}
homework	Array	Assignments: title, description, dueDate, classId, subjectId
homeworkSubmissions	Object	{homeworkId: {studentId: {submitted, grade, feedback}}}
teacherLeaveRequests	Array	Leave requests: status, substitutId, date range
studentHealthRecords	Object	{studentId: {allergies, medications, bloodType, ...}}

activityLogs	Array	Audit trail: userId, action, details, timestamp
archivedStudents	Array	Graduated or transferred students
schoolLogo	String	Base64 data URL or external URL of school logo
headmasterSignature	String	Base64 or URL of headmaster signature image
economeSignature	String	Base64 or URL of bursar signature image

6. Technical Architecture

6.1 Technology Stack

Layer	Technology	Purpose
Markup	HTML5	Single-file structure; all sections in DOM, toggled by JS
Styling	CSS3 custom properties	Earth-tone design system; responsive grid layouts
Logic	Vanilla JavaScript (ES5/ES6)	All CRUD operations, routing, computation
Charts	Chart.js (CDN)	Bar, pie, and line charts on the dashboard
Fonts	Google Fonts (Anybody + DM Sans)	Typography loaded from CDN
Storage	localStorage	All data persistence; no server required

6.2 Routing & Navigation

There is no URL router. Navigation is simulated by showing and hiding section elements. The **navigationConfig** object maps each role to an ordered list of {id, icon, label, section} entries. On login the sidebar is dynamically built from this config, and clicking a nav-link calls `navigateTo(sectionId)`, which hides all sections and reveals only the target.

6.3 Trimester Calculation

Two helper functions implement the Burkina Faso academic calendar:

- **getTrimestre(date)** — Returns {key, label, min, max} for the matching trimester, or null for July–September (school holiday period).
- **getCurrentTrimestre()** — Calls `getTrimestre` with today's date to auto-select the active trimester in forms.

T1: October–December | T2: January–March | T3: April–June

6.4 Average Computation

Subject averages are computed on the fly from raw test scores stored in `testScores[studentId][subjectId][trimester]`. Each entry holds a score and a `maxScore`. The formula normalises to a 20-point scale: **average** = $\Sigma(\text{score} / \text{maxScore} \times 20) / \text{count}$. The general average is then the arithmetic mean of all subject averages.

6.5 Chart.js Integration

Three charts are rendered on the admin dashboard:

- **classAveragesChart** — Bar chart: one bar per subject, Y-axis 0–20.
- **cepResultsChart** — Pie chart: admitted boys vs admitted girls; shows a placeholder message if no CEP data has been entered.
- **performanceTrendChart** — Dual-axis line chart: CEP pass rate (%) on the left axis and school average (/20) on the right, plotted over 2022–2024.

6.6 Data Persistence

Every state mutation calls `saveData()`, which serialises `appState` to `localStorage`. On page load, `loadData()` reads and merges the saved JSON back into `appState`, preserving backward compatibility for fields added in later development phases.

6.7 Security Considerations

Because all data lives in the browser's `localStorage`, the system is suited to single-device, trusted-environment use. Passwords are stored with minimal hashing. Activity logs provide a basic audit trail. For production deployment, a server-side backend with proper authentication and encrypted storage would be strongly recommended.

7. Development Phases

The codebase was built incrementally across four phases:

Phase	Focus	Key Additions
Phase 1	Core Customisation	School info settings, logo/signature uploads, subject management, fee configuration
Phase 2	Academic Core	Full grade entry (testScores), per-trimester averages, CEP results tracking, bus tracking
Phase 3	Communication & Enhanced Tracking	Instant messaging, announcements, teacher-class assignments, discipline records
Phase 4	Final Enhancements & Security	Homework & submissions, teacher leave requests, student health records, activity logs

8. Summary

École Numérique is a comprehensive, offline-capable school management system tailored to the Burkinabè educational context. In a single HTML file it delivers all of the following:

✓ Multi-role authentication — Admin, Teacher, Parent, Student	✓ Internal messaging and school announcements
✓ Complete student registration and class management	✓ Discipline and merit tracking
✓ Three-trimester grade entry, average computation, and report card generation	✓ Teacher leave management
✓ CEP examination tracking for CM2 students	✓ Student health record keeping
✓ Absence recording and monitoring	✓ Data export to CSV
✓ Timetable management	✓ Full configuration panel for school branding and identity
✓ School fees collection and exemption management	✓ Interactive Chart.js dashboards with live data
✓ Homework assignment and submission tracking	

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